

Aarav Kashyap

TOPICS OF INTEREST

Engineering, Computer Engineering, Robotics, Mechatronics

EDUCATION

Dublin High School, Dublin, CA

Class of 2026

- Coursework includes AP Seminar, AP Research, AP Statistics, AP Computer Science Principles, AP Computer Science Applications, AP Calculus AB, AP Chem, Honors Chem, Spanish 3, AP Calculus BC, AP Physics C Mechanics, Honors Digital Electronics, AP Environmental Science, Advanced Robotics Engineering
- Activities: Robotics - 5 years; Robotic Arm- 2 years

U.C. Berkeley Robotics Summer Program(Robotics Engineering), Berkeley, CA

2024

- Worked in a 6-week summer program on essential robotics concepts
- Participated in building robots, gaining exposure to robotics and the fundamentals of engineering
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U.C. Santa Cruz B.S. in Robotics Engineering

2026 - 2027

Boston University B.S. in Computer Engineering

2027 - 2030

WORK/VOLUNTEER EXPERIENCE

Gael Force Robotics Events Dublin, CA

2022 – present

Event Team volunteer

- **Robotics Club Volunteer:** Focused on STEM outreach, mentoring peers in coding and robot design, pushing through setbacks like build reiterations with determination
- **GirlPowered Event:** Helped the build station at our annual Robotics Club event, inspiring young girls in STEM with hands-on projects, and sharing my robotics passion to spark their curiosity.
- **Robotics Competition Host:** Co-manage our statewide competition, working daily on logistics—scheduling, setup, troubleshooting—creating a collaborative space for teams to compete for the State Championship spot.
- **Community Impact Leader:** Strengthen club outreach, adapting to challenges like technical difficulties or tight deadlines, never giving up, and growing my skills in leadership, teamwork, and problem-solving.
- Key Skills: leadership, time management, collaborating with peers, communication

OTHER EXPERIENCES

VEX Robotics Competition Team(5327C, 5327V, 5327V, 63174A)

2020 – present

- Served as team captain, overseeing strategy, team coordination, and project execution for competitive robotics challenges.
- Led the team with build, program, and testing robots for competitions, qualifying for the state/world championship.
- Contributed documentation for the robot's engineering design process, contributing to multiple awards.

Robotic Arm Project

2024 – present

- Designed and programmed a robotic arm controlled via keystrokes and EEG signals, integrating brain-computer interface technology to enhance functionality.

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- Independently developed and tested the system, showcasing innovation in robotics and human-machine interaction.

Dublin High School AI Club – Lead Research Sector

2023 – 2024

- Directed the research sector, teaching peers the fundamentals of artificial intelligence, including machine learning concepts, neural networks, and practical applications.
- Developed and delivered hands-on lessons to foster understanding of AI principles, enhancing club members' skills in coding and innovation.

School Club Accessibility App Development (Club Central)

2024 – Present

- Currently developing a mobile application to centralize information on all school clubs, making them more accessible to students.
- Designing features to boost student engagement with the school community by simplifying club discovery, participation, and communication.

Gael Force Robotics President

2025 - Present

- Got elected as President of Dublin High School's Robotics Club
- Club Consists of 100 - 200 members and is one of the biggest clubs in Dublin High School
- Expanding to Cottonwood Creek Elementary School and Emerald High School

Gael Force Robotics Summer Camp

2025 Summer

- Hosted a robotics summer camp for 3 weeks and taught elementary/middle school kids the fundamentals of robotics in the Bay Area
- made revenue of 2.5k+

Self - Research with Monte-Carlo Localization algorithm

2024-2025

- Conducted a research experiment with local high school robotics students with MCL algorithm
- was able to hypothesize a more accurate/precise programming algorithm used for VEX Robotics students

U.C. Berkeley ATDP Robotics Summer Program TA Internship

2025 Summer

- Helped teach robotics to ATDP's Advanced Robotics Engineering Globals Program
- Was able to help students from India and China learn the fundamentals of robotics/brainstorm new ideas

AWARDS/CERTIFICATIONS/TECHNICAL SKILLS

VEX Robotics Competition Achievements:

- Tournament Champion (2x, Locals)
- Excellence Award (4x, Locals)
- Create Award (1x, Locals; 1x, State Championship)
- Robot Skills Champion (2x, Locals)

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- Think Award (1x, Locals)
- Build Award (1x, Locals)
- Design Award (1x, Locals)
- Amaze Award (1x, State Championship)
- Innovate Award (1x, Locals)
- Top 10 Skills in the World (2x)

Skills

- **Programming Languages:** Fluent in Java, Python, C++, HTML, and Dart/Flutter, with extensive experience applying these in robotics, app development, and AI projects.
- **Hardware Proficiencies:** Proficient in breadboarding, soldering, and building printed circuit boards (PCBs), enabling the design and assembly of custom electronic systems of my robotic arm.